

Mediator in the Present: A Framework for Triadic Self-Dialogue in Mixed Reality

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Figure 1: Experience snapshot of the MR prototype: the present self mediates between a child self and a future self.

ABSTRACT

Many difficult decisions are conversations across time, between who one was, who one is, and who one might become. This position paper proposes *Mediator in the Present*, a triadic mixed reality (MR) self-dialogue framework that reframes reflection as embodied social interaction among co-present temporal selves. It externalizes temporal perspectives as avatars of a child self and a future self, and the present self as mediator, allowing users to inhabit, voice, and negotiate multiple viewpoints within the physical world. Building on prior work in embodied self-dialogue and future-self continuity, we outline design principles and interaction patterns for triadic temporal mediation, and demonstrate feasibility through a proof-of-concept MR prototype with avatar age transformation, age-shifted playback, and controlled embodiment alternation. Open research questions and ethical considerations are discussed for adaptive virtual humans and identity-relevant MR experiences.

Index Terms: Mixed Reality (MR), Triadic Interaction, Avatar

1 INTRODUCTION

Many difficult decisions are not arguments between logic and impulse, but conversations across time, between who one was, who one is, and who one might become. As immersive technologies and virtual humans are increasingly used to support reflection, self-understanding, and future-oriented decision-making [17], they offer a rare opportunity to make these inner conversations visible, inhabitable, and shareable through embodied experience. Prior work

shows that embodied self-dialogue in virtual reality (VR) can facilitate emotional regulation and perspective-taking [7, 14], while interactions with age-progressed future selves can strengthen future self-continuity and influence long-horizon choices [9, 15]. Both lines of research suggest that immersive interaction can turn temporal imagination into something felt, encountered, and negotiated rather than merely contemplated.

However, most existing systems frame self-reflection as a dyadic interaction between a present self and a single alternative perspective, such as a counsellor avatar or a future self. This dyadic framing provides a strong foundation for embodied self-reflection, yet it can simplify the reality of difficult decisions, which often involve tensions among multiple internal viewpoints shaped by past experiences, present constraints, and imagined futures. Such decisions are rarely resolved through advice alone; instead, they unfold through negotiation among competing values, emotions, and temporal priorities. Moreover, much of this work has been explored in VR, where reflective dialogue may feel less grounded in the physical environments where everyday decisions are lived and carried out.

Instead, mixed reality (MR) offers a unique opportunity to re-frame self-reflection as an intrapersonal social interaction situated in the real world. By externalizing multiple temporal selves as co-present virtual humans, MR can stage internal conflict as a structured, embodied triadic dialogue that supports sense-making rather than prescription. Rather than outsourcing reflection to an intelligent system, the goal is to amplify participant-authored reasoning by allowing the same idea to be expressed, witnessed, and negotiated across distinct temporal embodiments while remaining grounded in the surrounding environment.

To explore this idea, the paper introduces *Mediator in the Present*, a triadic MR self-dialogue framework in which a user engages simultaneously with an embodied child self and an age-progressed future self. The present self takes on the role of medi-

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ator, navigating affective memory, immediate needs, and long-term considerations through co-present interaction. In this framing, self-reflection becomes less a problem to be solved and more a dialogue to be held.

2 BACKGROUND AND INSPIRATION

2.1 Embodied Self-Dialogue in Extended Reality (XR)

A growing body of work demonstrates that embodied self-dialogue in immersive environments can support reflection, emotional regulation, and perspective-taking [4, 14, 7]. Early studies on embodied self-dialogue in VR showed that alternating embodiment between one's own avatar and a counselor avatar (e.g., a virtual Sigmund Freud) facilitated greater perceived insight and emotional change compared to scripted interactions alone [14]. Subsequent work extended this paradigm to intrapersonal contexts [18], enabling users to converse with different representations of themselves to practice emotion regulation or work through internal conflicts, with outcomes comparable to or exceeding traditional techniques such as the empty-chair method [20].

Related research has also explored the use of VR to reconstruct self-identity and resolve internal tensions in both clinical [7, 8] and non-clinical [12] settings, highlighting how embodied role-taking can externalize inner dialogue and make abstract psychological processes experientially accessible. These studies establish immersive self-dialogue as a viable mechanism for intrapersonal negotiation, but they typically rely on dyadic configurations and primarily use VR with few addressing MR or augmented reality (AR) for such embodied purposes [18].

2.2 Age-Progressed Avatars and Decision-Making

Research has also focused on interactions with age-progressed future selves to influence long-horizon reasoning and behavior. Seminal work demonstrated that embodying or interacting with a realistic future self in immersive environments increases preference for delayed rewards and future-oriented decisions [5], such as retirement saving [6]. Later studies showed that conversational exchanges with a future self can reduce self-defeating behaviors and increase future self-continuity across domains including financial decision-making, substance use recovery, and wellbeing [15, 13].

More recent work has begun to explore dialogue-based future-self interactions rather than passive visualization. Systems such as Future You frame future selves as conversational partners, demonstrating reductions in anxiety and increases in future self-continuity [11]. Research on age-incongruent avatars further suggests that interacting with a mature or older self can create a stronger immediate sense of perspective change, though effects are often short-lived and limited to dyadic exchanges [9].

Across this literature, future selves are typically positioned as advisors or outcome-oriented guides [5]. The role of past selves, particularly child versions representing affective memory and vulnerability, remains largely unexplored. Moreover, few systems frame future-oriented decision-making as a multi-perspective negotiation mediated by the present self.

2.3 Toward Multi-Perspective Self-Dialogue

Prior work shows that embodied self-dialogue and future-self interactions can support reflection and future-oriented decision-making, but most systems remain dyadic. Typically, users engage with a single alternative perspective rather than moderating or mediating among multiple viewpoints.

Table 1 summarizes key clusters of related work. Avatar-based self-dialogue and body swapping can increase perceived insight and support emotion regulation, while age-progressed future selves strengthen future self-continuity and influence long-horizon behavior. However, these approaches rarely model the co-present interaction of multiple temporal selves or examine how conflicting per-

spectives shaped by past experience, present constraints, and imagined futures are negotiated in real time.

This gap is especially salient for complex decisions, where internal conflict often reflects tensions across multiple temporal identities rather than a single present-future trade-off. The limited use of past-self representations and multi-party dialogue leaves open questions about how immersive systems might better support richer forms of intrapersonal negotiation.

3 TRIADIC MIXED-REALITY SELF-DIALOGUE FRAMEWORK

3.1 Framework Overview

In Mediator in the Present (Figure 1), the user is embodied as the present self and interacts with two age-incongruent self-avatars: a child self representing past-oriented affective memory and aspiration, and an age-progressed future self representing long-horizon consequence and continuity. These avatars are not autonomous conversational agents. Instead, they function as embodied dialogical positions that externalize distinct temporal standpoints of the same individual.

Interaction follows a record-and-replay procedure that intentionally prioritizes user-authored content over AI-generated dialogue. The user temporarily inhabits each temporal self to articulate arguments, concerns, and questions from that standpoint. These utterances are then replayed through the corresponding embodied avatars, allowing the present self to re-encounter them as co-present interlocutors. This design draws on dialogical theories of the self, which emphasize role-taking, externalization, and re-voicing as mechanisms for reflection and identity negotiation.

Although all dialogue content originates from the user, the interaction is not reducible to unstructured self-talk. Embodiment alternation, spatial co-presence, and mediated replay transform internal reasoning into socially situated interaction that can be observed, negotiated, and reinterpreted. The present self occupies the role of mediator rather than advice recipient, navigating tensions among values, emotions, and temporal priorities. By staging this negotiation as an embodied triadic interaction in MR, the framework supports reflective sense-making without prescribing outcomes or privileging any single temporal perspective.

3.2 Concept Design

Inspired by dialogical theories of the self and philosophical accounts of practical reasoning as value-based deliberation, the triadic structure is intentionally non-oppositional [1, 10]. The child self is not framed as impulsive or naive, but as a perspective grounded in affective memory, imagination, and aspirational values. The future self foregrounds consequence, continuity, and long-horizon coherence, articulating how present choices may shape lived experience over time.

The framework targets decisions marked by ambiguity and internal conflict, including those involving identity, wellbeing, relationships, and long-term commitments. By externalizing temporal viewpoints as co-present virtual humans, MR reframes self-reflection as embodied social reasoning situated in the physical environment. The goal is not consensus, but clarification of values, trade-offs, and priorities across time.

These avatars do not function as advisors or predictive models. Instead, each surfaces a distinct temporal standpoint, while the present self acts as moderator and mediator, negotiating tensions between aspiration, constraint, and consequence as they unfold through dialogue.

3.3 Implementation Pipeline (Proof-of-Concept)

To demonstrate the concept, a prototype¹ (see Figure 1) is implemented in Unity 2022.3.60 and runs on a Meta Quest 3 through

¹<https://github.com/shuwithu/TriadicMR>

Table 1: Summary of prior studies relevant to triadic mixed reality self-dialogue.

Core element	Evidence so far	Relevance to proposed framework	Key citations
VR self-counselling via body swap	Improves perceived change and emotion regulation	Demonstrates feasibility and benefits of embodied VR self-dialogue	[14, 20, 3, 7, 8]
Age-progressed and age-incongruent self-avatars	Increase saving behavior, immediate perceived change, and reduce self-defeating behavior	Validates using an older self perspective to support long-horizon reasoning and perceived insight	[6, 15, 13, 9, 11]
Future-self VR for decision conflicts	Present-future self advice exchanges guide choices	Similar format but dyadic, without past-self representation	[15, 9, 19]

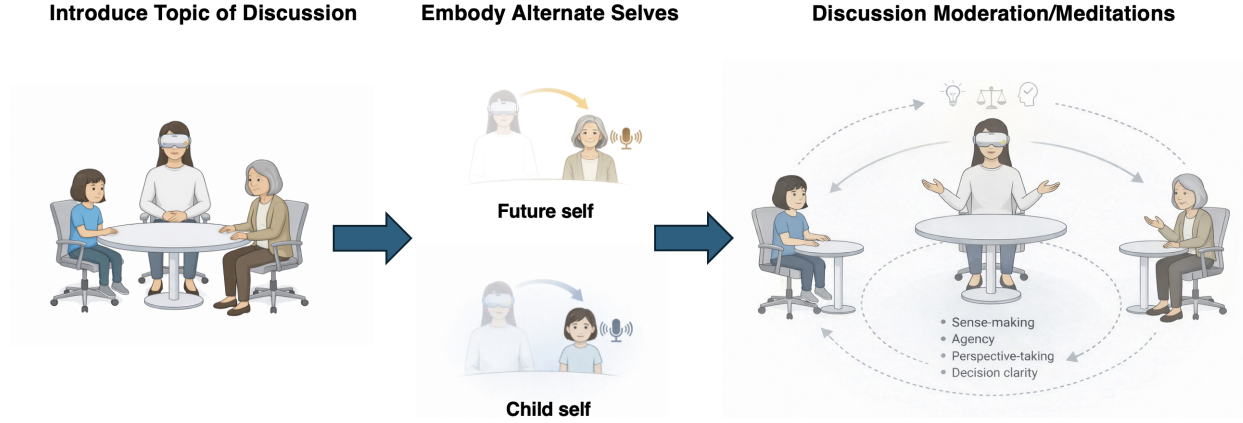


Figure 2: Framework structure and the paradigm procedure.

passthrough mode. Avatar creation begins with a high-resolution, front-facing portrait of the user. This image is used to generate two age-transformed versions of the user, one child (~ 10 years old) and one older adult (~ 60 years old). These generated reference images are then passed through the Character Creator² pipeline to produce corresponding 3D avatars for the triadic dialogue.

Perspective shifts are enabled through controlled embodiment alternation. This is implemented via first-person upper-body tracking, mapping the user's head and hand movements onto the embodied avatar during each perspective-taking segment. The avatars are spatially anchored using room-scanning with MR Utility Kit³ and can be repositioned or reconfigured at runtime. User speech is recorded in Unity, and age-shifted voice playback is generated via the ElevenLabs API⁴.

3.4 Further Expansion and Potential Add-ons

While the prototype demonstrates the feasibility of triadic MR self-dialogue, the framework is intentionally modular and can be extended towards more adaptive and socially grounded experiences. Several directions are promising:

- **Richer virtual human expressiveness** through affective design [16], such as dynamic facial animation, gaze, and turn-taking behaviors to strengthen co-presence and examine how embodied cues shape trust, engagement, and meaning-making in self-dialogue.

²<https://www.reallusion.com/character-creator/pipeline.html>

³<https://developers.meta.com/horizon/documentation/spatial-sdk/spatial-sdk-mruk/>

⁴<https://elevenlabs.io/>

- **Socially authored temporal selves** by allowing another person to embody one temporal self, or by grounding a temporal self in a mentor-like archetype, introducing interpersonal dynamics into intrapersonal negotiation.
- **Multimodal sensing for real-time adaptation** using lightweight behavioral and physiological signals (e.g., pauses, gaze shifts, heart rate) to modulate pacing, topic transitions, and emotional intensity.
- **Adaptive triadic dialogue with LLM support** to sustain distinct temporal voices across sessions and support iterative negotiation beyond replayed role-play, building on evidence that future-self conversations can shape affect and future self-continuity [11].

4 OPEN QUESTIONS AND ETHICAL CONSIDERATIONS

Triadic MR self-dialogue raises open research questions and opportunities for virtual humans, immersive interaction, and human-AI collaboration. A central question concerns how users negotiate credibility and emotional weight across multiple temporal selves. Prior work suggests future selves may be perceived as more consequential [2], yet it remains unclear how child and future perspectives interact when co-present, and how users reconcile conflicting viewpoints into a coherent decision narrative.

A second question concerns mediation dynamics. The framework positions the present self as an active mediator or moderator, but little is known about how embodiment alternation, spatial arrangement, and replay timing shape agency, ownership, and self-continuity. Even without generative dialogue, design choices such

as turn-taking structure, voice transformation, and avatar expressiveness may amplify certain perspectives and influence outcomes. Future work should examine when these mechanisms support reflection versus when they unintentionally steer interpretation.

Ethical considerations are particularly salient given the emotional intensity of past and future selves. Child-self embodiment may surface vulnerability and affective memory, while future selves may implicitly encode normative assumptions about success, health, or life trajectories. The framework therefore emphasizes user-authored content, transparency about any AI-mediated transformations, and support for multiple plausible futures rather than a single optimized path.

Finally, triadic self-dialogue raises questions about boundaries and responsibility in immersive self-augmentation. As virtual humans become embedded in identity-relevant experiences, it becomes essential to distinguish systems that scaffold sense-making from those that exert undue influence. Addressing these challenges will require interdisciplinary work across HCI, psychology, and ethics [17, 4], positioning triadic MR self-dialogue as a valuable testbed for responsible reflective interaction.

5 DISCUSSION AND IMPLICATIONS

This work frames triadic MR self-dialogue as a form of social interaction, extending collaboration beyond multiple users to include embodied temporal perspectives of the self. By positioning the present self as a mediator between child and future selves, it illustrates how immersive environments can support negotiation, perspective-taking, and sense-making during reflective decision-making. In this view, social interaction in XR can arise not only from co-located people, but also from structured dialogue among co-present virtual humans that represent internal viewpoints.

Importantly, neither avatar is treated as authoritative or correct. Each offers partial and situated thoughts, while the present self retains full agency over interpretation and choice. It treats reflection as negotiation rather than optimization, emphasizing value clarification and meaning-making over prescriptive guidance.

The framework also suggests practical directions for human-AI interaction design in XR. Virtual humans can be designed as relational participants with distinct temporal roles, rather than as standalone assistants. Embodiment alternation and mediation provide reusable interaction patterns to support reflection without forcing outcomes. While this paper presents a proof-of-concept implementation, the framework is intentionally modular, enabling future work to explore different temporal roles, visuals, and dialogue structures across domains (e.g., career, health, relationship).

More broadly, triadic self-dialogue offers a testbed for studying identity-related interaction over time in XR, including its ethical boundaries. By externalizing temporal perspectives as co-present virtual humans, this framework makes internal negotiation observable, discussable, and designable. In the end, the hardest choices may not be solved by better answers, but by learning to mediate between the selves that each carry a piece of the truth.

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